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Developing a Knowledge Management System for Consulting Firms

A Framework for Knowledge Sharing in Project-Driven Product Development

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Utveckling av ett kunskapshanteringssystem inom konsultföretag: Ett ramverk för kunskapsdelning inom projektbaserad produktutveckling

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Sammanfattning

Dagens snabbt skiftande och konkurrensutsatta marknad har lett till att organisationer blivit alltmer kunskapsintensiva. Detta ställer krav på organisationers kunskapshantering. För konsultföretag är detta särskilt viktigt, med tanke på att de bygger sin verksamhet på den tillgängliga kunskapen. Till följd av alltmer kreativa och drivna konsulter så sätts nya utmaningar på hur hanteringen av kunskap ska ske.

Syftet med denna studie är att utveckla ett system för kunskapshantering hos konsultföretag med fokus på projektdriven produktutveckling, med målet att presentera ett förslag på ett tillämpat system för inContext AB. För att uppfylla detta syfte så definierades viktiga begrepp som används i denna rapport. Sedan delades arbetet upp i en infrastrukturell utvärdering, en analytisk fas, en designfas samt en utvecklingsfas.

Metoderna som användes i den infrastrukturella utvärderingen och analytiska fasen var baserade på abduktiv forskning. I den abduktiva forskningen kombinerades en kvalitativ studie som inkluderade intervjuer, observationer och informella samtal tillsammans med en litteraturstudie. Avsikten var att skapa en god förståelse av kunskapshantering och dess utmaningar. Metoderna som användes i design- och utvecklingsfasen baserades på vetenskapliga ramverk, modeller och metoder för att på ett logiskt sätt kunna designa och utveckla kunskapshanteringssystemet.

Resultat visar vilka kritiska framgångsfaktorer som är relevanta för inContext AB i synnerhet och konsultföretag i allmänhet. Resultatet visar även vilka nyckelfunktioner för kunskapshantering som redan finns på plats såsom sammankomster, mentorskap och Google Drive, samt identifierade utmaningar där nya nyckelfunktioner som utlärningssystem och konsultprofiler bör implementeras. Det utvecklade systemet för kunskapshantering beskriver nyckelfunktioner som bör implementeras för att säkerställa en god kunskapshantering med hänsyn till de utmaningar som finns, men även nuvarande relevanta nyckelfunktioner som bör behållas och vidareutvecklas.

Slutsatserna är att alla nyckelfunktioner i det utvecklade systemet för kunskapshantering skulle vara fördelaktiga att implementera. En kritisk avvägning är dock att alla identifierade nyckelfunktioner inte kan implementeras på en och samma gång. Detta på grund av budget, ansträngning och implementeringstid. Därför gjordes prioriteringarna att standardisering av befintliga nyckelfunktioner skulle vara första prioritet. Därefter rekommenderas det att lägga fokus på att implementera de nyckelfunktioner som bygger och utvecklar befintliga processer. Som ett

sista steg rekommenderas implementering av de nya nyckelfunktioner som förväntas komma att främja kunskapshanteringen för inContext AB framgent.



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Abstract

Organizations are becoming more knowledge-intensive in today's competitive environment. This is important to consider for consultancy firms that are building their business on knowledge as their key resource. Because of the increasingly educated and creative workforce, this shift introduces new challenges in managing knowledge.

The purpose of this master's thesis is to develop a system for knowledge management in consultancy firms with focus on project-driven product development. It aims to propose an applicable system that can be implemented effectively for inContext AB. To fulfill this purpose important topics were first described in the frame of reference to give a good understanding of terminologies used by the scientific community in regard to knowledge management. Then the work was divided into four phases, infrastructural evaluation, analysis, design, and development phase.

The methods used in the infrastructural evaluation and analysis phase were based on abductive research. Combining a qualitative study, which included interviews, observations, and informal conversations, with a literature study. With the intent of creating a body of knowledge and understanding the challenges. The methods used in the design and development phase were based on using scientific frameworks, models, and methods in a logical manner to design and develop the knowledge management system.

The results show which critical success factors are relevant for inContext AB in particular and consultancy firms in general. It also shows which key functions are already in place and are affecting knowledge management in a positive manner such as gatherings, mentorships, and Google Drive, as well as challenges where new key functions like learning management systems and consultancy profiles should be implemented. The developed knowledge management systems describe key functions that should be implemented to ensure good knowledge management regarding the challenges found, as well as good key functions that should be kept and further developed.

The conclusions were drawn that all key functions in the developed knowledge management system would be beneficial to implement. Although due to time, effort, and budget, all key functions cannot be implemented simultaneously. Therefore, prioritizations were made that standardizing existing key functions should be the first priority. Thereafter, the focus should be on

implementing key functions that build on and develop existing processes. Lastly, implementing new key functions that can contribute to solving the challenges that inContext AB faces.

FOREWORD

In this chapter, we would like to acknowledge the people who helped and supported us during this study.

First and foremost, we would like to thank our supervisor Johan Bergström at inContext AB for providing assistance and input to the project.

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Joel Bergqvist and Carl Lindell

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NOMENCLATURE

Abbreviations

CKO	Chief Knowledge Officer
CSF	Critical Success Factor
ICT	Information and Communication Technology
KM	Knowledge Management
KMS	Knowledge Management Systems
KPI	Key Performance Indicator
LMS	Learning Management System
ROI	Return on Investment
SKMI	Systematic Knowledge Management Initiative

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1 INTRODUCTION

In today's competitive climate it becomes more common for organizations to transform into knowledge-intensive structures. This has flourished the landscape of consultancy firms which build their business on knowledge and has knowledge as the key resource instead of labor, raw materials, or capital. This organizational offering of complex problem-solving services sets trends which changes the conditions for consultancy firms. Most of the work force becomes well-educated, creative, and driven. This also sets new challenges regarding how an organization manages its knowledge both in terms of explicit and tacit knowledge.

1.1 Background

This master's thesis has been performed in collaboration with inContext AB, a consultancy firm specializing in complex electrical systems. The head office is located in Stockholm. Recently, they opened a new office in Gothenburg, a strategic location due to the increased electrification industry in the area. Today, they offer expertise in electrical equipment, complete electrical systems, mechatronics design, and project management and are running projects both on-site and in-house within different industries, such as the automotive industry, truck industry, boat industry, and military industry. Focusing on cable harnessing is a niche approach they have taken recently, they still have ongoing projects in areas as mechanical engineering and software development. However, the management group has decided that moving forward, the company will be more decisive about the projects they choose and to be more focused on electrical systems. At the start of 2023, inContext AB had around 80 employees, which puts them in the bracket of a smaller midsize company. They have a high focus on making sure that projects and assignments meet the targets and deliver on time to maintain good relations with the customers. Currently, their market is lucrative and is scaling at a rapid pace in terms of both revenue and employment. This has sparked further interest at the management level in understanding more about knowledge management since it's an area that has become more challenging. It is also common knowledge within the research community that knowledge management becomes harder to manage the bigger an organization gets. As top management sees it, the current problem area is that their current system for knowledge management is lacking both in terms of structure and keeping it up-to-date. They also see the problem of being able to teach someone a program or work method for problem solving but not being able to demonstrate competence in that program. The difference there is described as having knowledge of the functions of how a program works, contrary to being able to understand and solve problems with a program.

1.2 Purpose

The purpose of this master's thesis is to develop a system for knowledge management in consultancy firms with focus on project-driven product development. It aims to propose an applicable system that can be implemented effectively for inContext AB.

Research question 1:

Which critical factors preventing respectively encouraging knowledge sharing in connection to product development projects can be identified?

Research question 2:

What are the key functions in a knowledge management system for product development businesses and which critical trade-offs are needed when developing such a system?

1.3 Delimitations

The study was limited to 30 credits (ECTS), which correspond to 20 weeks.

Although this study will research current capabilities and needs revolving the technology dimension, it will not develop any new components for a knowledge management system, due to time constraints. Furthermore, due to the limited time for implementation of a knowledge management system, neither the effects nor the cost estimation of the developed framework will be performed.

2 FRAME OF REFERENCE

In this chapter explanations are given to important definitions used in this project. These definitions are important to understand to be able to recognize what this study is researching. Also, to get a deeper understanding of why later chosen methods are appropriate to use. And lastly to realize what this study is intending to accomplish as well as empathize with the results.

2.1 Definition of knowledge

Knowledge refers to something that is learned through experience. The terms information and knowledge are occasionally used interchangeably. The truth is that these two terms are not synonymous. Information can be thought of as refined data, whereas knowledge is the application of relevant information. Furthermore, knowledge is a level of comprehension that goes beyond information, necessitating the ability to apply information meaningfully, draw conclusions, and produce new ideas and perspectives. Additionally, knowledge can be classified into explicit and tacit knowledge, which are further described in the upcoming chapter. (Surbhi S, 2018)

2.1.1 Explicit and tacit knowledge

Explicit and tacit knowledge are terms presented by Nonaka and Takeuchi (1995) to describe different kinds of knowledge. Explicit knowledge can be described as knowledge that can be shared through manuals and theories. Also, it can be seen as rational and objective. It is also created sequentially and is oriented towards a theory about something. Tacit knowledge is the opposite, which means something you can't express in a manual and is best expressed and learned through experience. It is much more subjective and oriented towards a person's beliefs and values. Therefore, to transfer this type of knowledge, a more analog process is required, where learning-by-doing practices serve as an example.

2.2 Knowledge management

Knowledge management (KM) is a process for capturing and distributing knowledge within an organization (Maier, 2007). It involves processes for identifying and capturing important information, creating a culture where you are willing to share your expertise, and how to make it easily accessible. The goal of knowledge management is to work smarter, not harder, by creating a more efficient and effective organization by leveraging the collective knowledge and expertise of its members. Knowledge is the most valuable resource in consultancy firms, so it must be considered important. Awad and Ghaziri (2007) describe it as if you are a part of the management group at the company, you can ask yourself questions like: Do you know what your employees know? Do you know what you know? How do you make the best use of the knowledge possessed within the company?

2.2.1 Critical success factors for knowledge management

For knowledge to be managed effectively there are numerous critical success factors (CSF) for knowledge management that need to be considered. Several studies have been conducted on CSF and in a literature review by Onofre and Teixeira (2022a) a total of 116 critical factors were identified that impact knowledge management. Instead of identifying all factors from scratch, this review was used as a basis for which CFS to considered and complementary articles were studied to see if all CFS that should be considered were included. Instead of presenting and explaining all

CFS in the frame of reference the reader is referred to Appendix B:4:. This is done to keep the frame of reference at a reasonable length and promote easier readability for a more applicable interested reader. The presented CFS need to be handled for effective knowledge management and one way of handling CFS is establishing a knowledge management system (KMS).

2.2.1 Knowledge management systems

Knowledge management systems can be described to fall under two strategies, push and pull. The two systems could hence be described in two different ways. Push strategy adopts a more technology-based approach (Andrea Di Stefano, 2022). Therefore, a push KMS could be described as a technology-supported information system that helps the organization to document, distribute, and transfer knowledge between employees to increase organizational effectiveness (Yang et al., 2014). A pull KMS can be considered as a more social way of sharing knowledge. Focusing more on participants actively searching for knowledge and experts offering it. Reeling less on technology systems and more on creating knowledge networks of people (Andrea Di Stefano, 2022). A KMS helps ensure that valuable information is not lost when consultants leave or retire, as a result of supporting knowledge sharing and coping with knowledge hoarding. It also facilitates collaboration among team members and projects, which can lead to better decision-making. Continuously, it fosters a culture of continuous improvement by promoting the power of sharing and lessons learned across the organization, which in turn enhances development opportunities for employees. When designing a KMS it is proclaimed throughout the academic community that four pillars need to be considered. These pillars are phrased differently but usually encapsulates the same areas. For this project the decision was made to use organization, process, strategy, and system as the four pillars for a KMS, which can be seen in Figure 1 below.

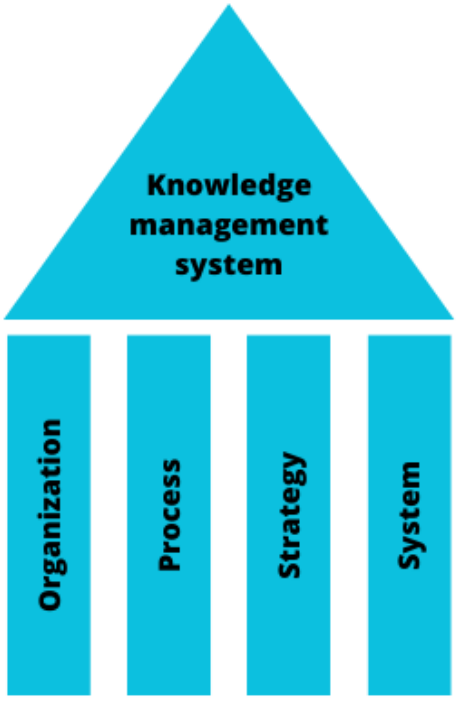


Figure 1. Four pillars of a knowledge management system.

The pillars are defined as follows:

- **Organization**: The organizational pillar includes the structure and culture that supports knowledge management. This includes creating a culture that values knowledge and

encourages knowledge capturing, storing, and sharing as well as fostering a collaborative work environment. This pillar will also include the human resources aspect of knowledge management. Which means the identification, recruitment, development, and retention of knowledgeable individuals within an organization.

- **Process**: The process pillar includes the various processes and procedures used to manage knowledge within an organization. This includes identifying critical knowledge areas, maintaining knowledge repositories as well as establishing procedures for knowledge capturing, sharing, and storing. Processes for quality control measures to ensure the accuracy and relevance of knowledge is also included.
- **Strategy**: The strategy pillar refers to a plan or approach to accomplish specific long-term objectives of the company. Strategies are frequently used to guide decision-making and resource allocation within the company. These strategies are often established at an organizational level and include a broader view compared to processes.
- **System**: The system pillar refers to the technological tools that a company uses to enhance knowledge management. This includes communication and collaboration tools, as well as technologies that aid in the collection, development, and sharing of knowledge.

When compared to other systems, KMS is more collaborative and participatory, which also sets some requirements for systematic initiatives. These can be considered as support for the KMS to stand.

2.2.2 Systematic knowledge management initiatives

Systematic knowledge management initiatives (SKMI) are initiatives surrounding KMS in order for them to work effectively and efficiently (Maier, 2007). These initiatives can include KM strategies and development of KM related goals. Creating a culture that supports KM like assigning roles designated to perform knowledge-related tasks and using processes that utilize KMS. Also developing corresponding measurement which evaluates whether the KM related goals have been achieved. There is more awareness for this part of KMS and therefore research has included this concept to a much greater extent in the management part of KMS. Even though the definition of KMS includes SKMI, SKMI will still be used for further distinction and to describe a specific part of KMS.

2.2.3 The SECI model

The SECI model is a framework for knowledge management developed by Nonaka and Takeuchi (1995). The model describes the process of looping tacit information into explicit knowledge and back again in order to generate new knowledge inside an organization. It incorporates an interaction dynamic in which knowledge is shared in a loop, increasing the value of knowledge through exchange between individuals, groups, and within the organization (Rice & Rice, 2002). A visualization of the model can be seen in Figure 2 below. It consists of following four components:

- **Socialization**: which is the process of passing on tacit information through direct human experience, such as mentorship or on-the-job training. This is seen as the place where individuals can share feelings, experiences, and perceptual models. (Rice & Rice, 2002)
- **Externalization**: which is the process of converting tacit knowledge into explicit knowledge. Documentation, training materials, and similar forms of communication are commonly used to accomplish this. (Rice & Rice, 2002)

- **Combination**: which is the process of combining explicit knowledge to generate additional explicit knowledge. This is accomplished through methods such as data mining, benchmarking, and exchanging best practices. (Rice & Rice, 2002)
- **Internalization**: which is the process of combining explicit knowledge into tacit knowledge. Experimentation, reflection, and practice are frequently used to achieve this. (Rice & Rice, 2002)

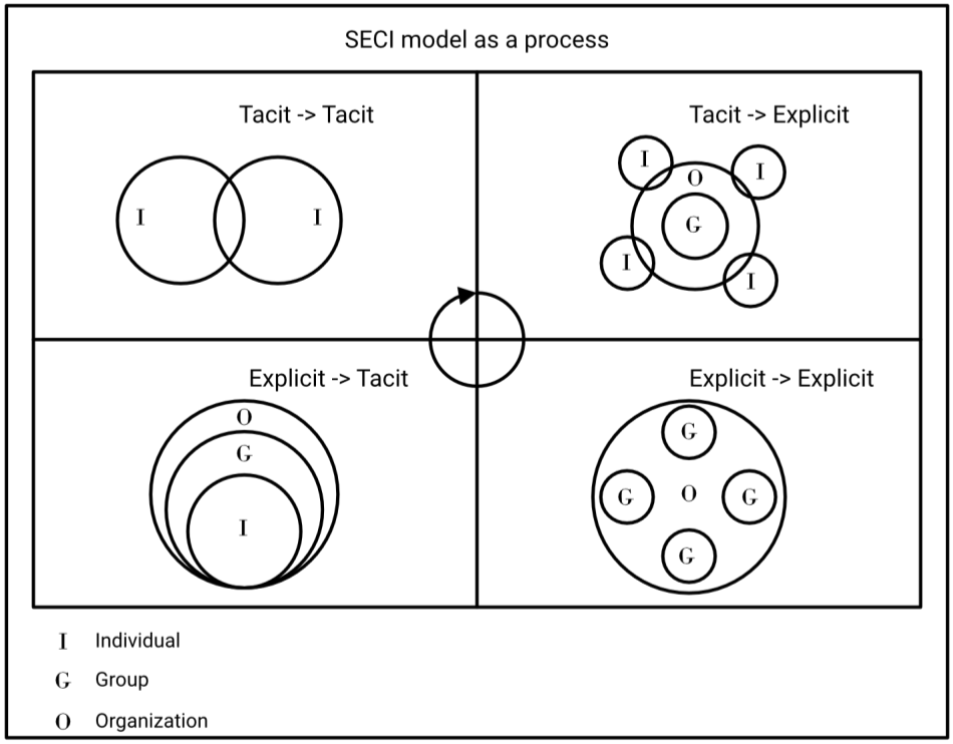


Figure 2. Elements of the SECI model and knowledge exchange between different segments.

This model is a well-known framework for comprehending the dynamic concept of knowledge management, which makes it suitable for observing KMS assistance for tacit knowledge transformation. Arguably, IT cannot replace humans when processing, making decisions, or acting on tacit knowledge. However, it can instead assist tacit data-based decisions into explicit knowledge alteration and vice versa, as well as tacit-oriented knowledge processes. Therefore, the SECI model helps facilitate both explicit and tacit knowledge. (Lesjak & Natek, 2021)

3 METHOD

The work was divided into four phases: infrastructural evaluation, analysis phase, design phase, and development phase. Each phase was further broken down into smaller activities based on methodologies used, previous knowledge, and personal experience. The coming chapter will describe what was included in each phase as well as detailed descriptions of the activities performed.

3.1 Infrastructural evaluation

To obtain a broad insight into the organization to be able to conduct an infrastructure evaluation of their current knowledge management capabilities, different methods were used. First, introductions were held by consultants from different projects and positions. This was an opportunity offered by HR, and new employees undergo a similar experience. The introduction consisted of employees introducing themselves, their backgrounds, their project team, their current project, and working methods. Additional details on ways of working were requested specifically for this project as well. All information for the infrastructural evaluation was gathered by taking notes and writing down reflections. Secondly, informal conversations were held with people from the management group. The conversations revolved around topics such as history of the company, general information, ways of working, and projects. Relevant information was documented in notes.

3.2 Analysis phase

The general approach for the analytical phase was based on abductive research and it was performed to find the answer to the first research question. Abductive research can be explained simply as making observations and seeking the hypothesis that best suits the observation. Abductive research therefore begins by making observations of an area and set of unrelenting facts. From there with intuition or reasons a hypothesis is formulated which seem plausible for the observations. Lastly the hypothesis is validated using other observed facts or studies. (Gordon, 2022)

The approach involved conducting a combination of observation and interviews in order to further develop hypotheses that could then be compared and verified by a literature study and vice versa. Both the literature study and the interviews are further described in detail below. This process was back and forth as observations, interviews and literature studies were done simultaneously over a period of time. During the whole duration of the project, attendance at the company was also conducted on a regular basis. This also accounted for a part of the abductive research since it was performed to get insightful, informal conversations, and observations that produce key information relevant to the project.

The approach for observing was based on phenomenon-based research. Phenomenon-based research is a qualitative research approach that focuses on the study of a phenomenon as it occurs in real-life situations, rather than in artificial or controlled environments. According to von Krogh et al. (2012), the aim of phenomenon-based research is to capture, describe, document, and conceptualize a phenomenon so that appropriate theorizing and the development of research designs can proceed. The course of action of this approach can be divided into five different research activities. Those activities are to distinguish, explore, design, theorize, and synthesize.

The phenomenon being studied can be anything from a social issue, a behavior, or a natural occurrence. Undoubtedly it has different features, depending on the scientific field into which they

fall and may increase in complexity and decrease in generality. Phenomenon-based research can never replace traditional theory-building and testing, it can complement that approach. It is valuable since it gains a deeper understanding of a specific phenomenon by focusing on the perspectives and experiences of those who experience the phenomenon. Further, when many new phenomena are discovered, knowledge creation increases. (von Krogh et al., 2012)

3.2.1 Literature study

In more detail, an extensive literature study was conducted in an early stage of the project. The purpose of the literature study was to firstly identify the current state of knowledge on the topic to get a solid foundation of existing literature to build on. The goal was to broaden insights on the topic and identify themes, concepts, and theories as well as methodologies and approaches used by other researchers when studying knowledge management. Another goal was to create an understanding of the topic extensive enough to formulate relevant research questions and place our research in an adequate context of the larger research domain of knowledge management. With the intent of ensuring that our research contributes to the advancement of knowledge within the field of study. To ensure our research encapsulated as a large area of the field as possible the following research terms in Table 1 below were used. Articles were then chosen by skimming through abstract, method, results, and discussion and assessing based on relevancy, credibility, authority, and objectivity.

Table 1. Research terms used for the literature study.

Critical factors for knowledge management
Factors affecting knowledge management
Knowledge centered culture
Knowledge management
Knowledge management in consultancy firms
Knowledge management systems
Measuring knowledge management
Motivation - Apter reversal theory
Risk taking and knowledge management
Social knowledge management processes
Tacit, implicit, and explicit knowledge
Technology brokering
Trust impact on knowledge sharing
User-satisfaction-based knowledge measurement

The second purpose of the literature study was to identify critical factors affecting knowledge management. This was done by reviewing all relevant sources chosen from the mentioned search

topics and documenting each relevant dimension, category, and factor. To further identify gaps in the existing research correlating to consultancy firms and to base suggestions for a system applicable to inContext AB on tested theories. The majority of all sources, around 20, were used to fulfil these first two purposes. Supplementary literature studies, in smaller scale, were undertaken throughout the project to help with decision-making and to confirm ideas with previous research and information obtained from the qualitative study.

3.2.2 Qualitative study

From the infrastructural evaluation and literature study a broad body of knowledge was obtained about both the topic of knowledge management and the organization inContext AB. Based on this an interview guide was constructed to ask open-end questions which could reveal challenges in inContext ABs knowledge management. Interviews asking specific questions about all 116 factors could not be held. Therefore, the questions were formulated with the intent that the respondent could answer with several different factors to the same question. A first draft of the interview guide was presented to the supervisors. This was made to get their input on how suitable the questions are and how they are interpreted. After some adjustments of the draft, a final version of the interview guide was created. The interview guide can be found in Appendix A. The semi-structured interviews were conducted with 10 respondents ranging in positions from top management to team leaders, to senior and junior roles. The interviews were conducted both on-site and digitally, depending on the respondent's availability. Both thesis workers were present at all interviews, where one held the interview and the other one was responsible for taking notes and preparing follow-up questions where more details were of interest.

3.3 Design phase

From the qualitative study a deeper understanding was obtained of inContext AB knowledge management capabilities and challenges. In the design phase inContext ABs challenges and capabilities were correlated to critical factors. Factors considered not critical were deducted. The following part goes over the process of deducting factors in a systematic manner.

3.3.1 Systematic deduction of factors

The systematic manner of deducting factors is illustrated in Figure 3 below.

The evaluation stage was done by first summarizing all factors in a document keeping the same structure with dimensions, categories, and factors as the literature review by Onofre and Teixeira (2022a). Deduction of factors was then performed by color coding and carried out in the following steps:

- First, dimensions were eliminated which had no critical influence for this specific case study. Here, red was used to indicate which factors had been eliminated.
- Secondly, categories were eliminated based on the two statements; capabilities are currently in place to manage the critical category; the category can be considered not critical for this case study. If either statement could be supported by the qualitative study, literature support or logical reasoning the category was eliminated. Here, orange was used to indicate which category had been eliminated.
- Thirdly, factors were eliminated based on the two statements; capabilities are currently in place to manage the critical factor; the factor can be considered not critical for this case study. If either statement could be supported by the qualitative study, literature support or logical reasoning the category was eliminated. Here, yellow was used to indicate which factor had been eliminated.

The remaining factors were then sorted into the four pillars of knowledge management that were chosen to be used in the beginning of this project.

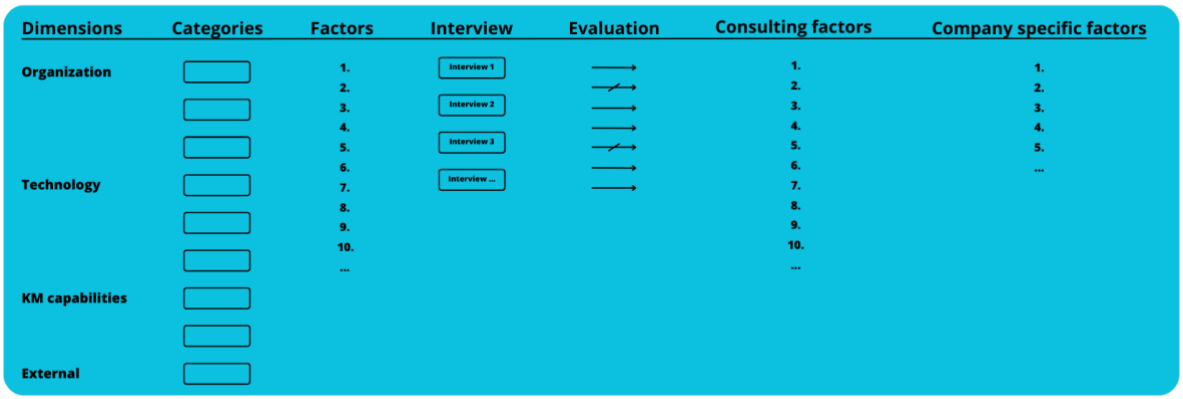


Figure 3. Illustration of the method used for deduction of factors.

3.4 Development phase

Development of the finalized KMS was done by combining the four pillars of knowledge management with the SECI model and complementing with the extra column of SKMIs as seen in Table 2. The headliners were changed to make it more general understandable. Therefore, socialization was exchanged by knowledge sharing, externalization by enable sharing, combination by documentation, internalization by learning, and lastly SKMI was exchanged by incentives. The framework was thereafter used by sorting in all current knowledge management, assessing where key functions were lacking and suggesting new ones to promote all types of knowledge transfer with different pillars.

Table 2. Template for the framework.

	Social sharing	Enable sharing	Documentation	Learning	Incentives
Organization	Key function 1				
Process	Key function 2				
Strategy					
System					

3.5 Validation

To finish off the qualitative study, an open discussion session was held about the KMS developed in *3.4 Development phase*. The session was held with the management group from inContext AB, since that was the group of individuals who had the most insights into the topic, but more importantly the organization which the proposed key functions would affect. This led to an even deeper discussion of the key functions and based on the thoughts and feedback minor adjustments were made to the KMS.

4 RESULTS

The following chapter presents the results for each phase in the project. The results are divided into infrastructural evaluation, literature analysis, results from the interviews, design of the KMS, and validation.

4.1 Infrastructural evaluation results

From the infrastructural evaluation, it has been found that inContext AB currently has no defined KMS or SKMI in place. However, this does not entail that there is no knowledge management being performed in the organization. There are logical procedures and what could be called knowledge management subsystems and systematic knowledge management initiatives put in place, but without a holistic view and the goal of creating a well-developed KMS with correlated SKMI. These procedures, initiatives, and subsystems also handle explicit and tacit knowledge without the intent being so from the beginning. Procedures, initiatives, and systems identified as already in place were:

A mentor is assigned to new consultants often being someone within the project the consultant has been assigned to. The mentor has the responsibility to help the new consultant in any way possible to get them independently working within the company. This mentorship is designed so that the first two months the mentor should in their work tasks allocate time to help the new recruited. After the first two months this is no longer a requirement from management, but the relationship often extends naturally after two months according to the management.

Training possibilities are offered to consultants. These trainings could either relate directly to current working areas or be courses to explore new fields within the electric transformation, cardiopulmonary resuscitation, and similar. inContext AB has the goal of providing a certain number of courses each year and for 2022 they reach around 80% of their goal. For 2023 they have also increased the goal of how many educational training possibilities they hope to achieve. One with a team lead position stated that there is a strategy for choosing employees who get to go on educational training. Decisions are made to an extent based on personal traits like willingness to learn, how good you are at sharing knowledge with others, and how well you can teach.

To try new roles within projects are both relatively easy to do, encouraged and help is seemingly easy to get. Although there seems to be a more frequent response from those who are either part of a larger group from inContext AB or who are running projects for a company they have been working with for a long time. To try new roles between projects also seems like a good possibility since management does not only delegate based on expertise but also have a dialogue with each consultant and tries to meet their wants.

On a yearly basis, management sits with each consultant and discusses personal development. For this meeting there is a standardized personal development plan with five dimensions: experience, knowledge, management, independence, and inContext AB, which are assessed. Each dimension can be correlated with knowledge management since there are several statements in each encouraging seeking or sharing explicit or tacit knowledge. The argument can be made with this being their core strategy for evaluation that inContext AB is to a great extent a knowledge centered organization. Even though this being the case the personal development plan was not brought up until the sixth introduction even though open-ended follow up questions about how management encouraged knowledge sharing were asked.

On several occasions each year, often on pay days, inContext AB arranges gatherings. These gatherings provide an opportunity for the consultants to socialize and interact with each other in a more open setting. It also allows consultants from various departments or teams to connect and exchange knowledge and experience. Furthermore, it expands the possibilities for engagement by demonstrating that inContext AB values its consultants beyond their work contributions. Increased engagement would result in more productive, motivated, and loyal consultants for inContext AB, boosting the likelihood of successful knowledge management. Gatherings can therefore foster a knowledge-sharing environment at the company.

inContext AB uses a collaborative platform called Workplace where events, news, gatherings, and social updates about the company are posted. The consistency of updates and attendance at events are high, which has led to a strong culture and profound relationships between coworkers. These can be considered strong SKMIs for creating openness, collaboration, and trust between coworkers.

Team leaders at inContext AB are getting introduced to the organization's guidelines. These guidelines provide reasons why to adopt them, what to do, and how to perform them. The scope of these guidelines is to have a standardized work setup and process for the projects, they are also relevant for knowledge management. An organization where knowledge management works well goes hand in hand with having the correct soft skills. Soft skills that might have a connection with knowledge management are following:

- *Be empathetic*
  To feel and understand other persons is important to gain the trust within the team. Further, trust will strengthen the relationships between the team members which in turn leads to better cooperation and engagement. The result of being empathetic improves the team productivity.
- *Communication ability*
  Having an environment where interactions in person can be easily done on a daily basis will increase efficiency and quality. Seeing that the person you're speaking with is taking in what you're saying encourages you to share your expertise and thoughts.
- *Empower the team members*
  Each team member can contribute to make the team better. Also, to work cooperatively and effectively with others to achieve common goals. This makes for better engagement and motivation.

Some team managers work with these soft skills within project groups to try and establish a good culture within the team. Although, the choice to implement this strategy comes down to personal decisions made by that specific team lead and it's not a standardized process.

Some team leaders work with quality assurance before releasing project results to clients, meaning that the work of more junior consultants is reviewed for possible improvements. If anything is found, the team leader revamps it together with the creator of the work. This is to establish improvements and learning experiences for future work. Although, again, the choice to implement this strategy comes down to personal decisions made by that specific team lead, it's not a standardized process.

HR works with company culture in general and touches on more topics than just knowledge management. One part of that work is conducting weekly surveys where employees give their opinions on certain matters. During the time this project was conducted, knowledge-

related questions were incorporated to some extent since this is a subject they aspire to improve.

4.2 Literature analysis

The CSF can be viewed in Appendix B:4:. The four dimensions which all factors can be sorted into are explained briefly below for a holistic view of the CSF included.

Organization: The organizational dimension includes how the organization operates and interacts with its environment. It includes areas such as structure, culture, strategy, and management.

Technology: The technology dimensions include, as the name infers, all IT-related structures like information and communication technology (ICT), which are used in KMS. The dimension also revolves around the application of these systems in terms of effectiveness, balance between IT and people, user friendliness, security of system infrastructure, etc.

Knowledge and KM capabilities: The knowledge and KM capabilities dimension focuses on the structure that is used to handle knowledge. This includes processes and procedures for how knowledge is created, captured, stored, shared, and applied. This dimension revolves around handling the efficiency of each KM process.

External influence: The external influence dimension focuses on factors that are beyond the organization's control. This allows for mapping out strengths and weaknesses. Influencing factors could be socioeconomic, political, industrial, and environmental.

Additional critical factors found not mentioned in the review by Onofre and Teixeira (2022a) are listed below.

Consultant turnover: The frequency in which consultants leave an organization is referred to as consultant turnover. This indicator, which is usually stated as a percentage, is used to track the number of consultants who leave an organization over a specific period of time. A high consultant turnover may include some negative effects, such as decreased productivity and increased hiring and training costs. Reasons for why consultants may leave an organization could be low satisfaction, a lack of progress, or poor management. (Ben-Dror, 1994)

4.3 Results from the interviews

The collected data was gathered from a total of 10 interviewees with consultants at inContext AB. Due to the fact that the organization does not have an articulated KM plan, there is naturally room for improvement. The results below display a selection of quotes that indicate critical factors in connection to knowledge management.

Make use of the knowledge bank

A majority of the respondents stated that there was a knowledge bank, and this seems to be general knowledge in the organization although that is the case all respondents also state that it's not updated, and therefore not used to its full potential.

"During all my years at inContext AB, we have tried to save various guidelines and manuals and things like that. But then new ones join and then nobody informs where to find it, and then it's just a big black hole." - Respondent F

Importance of building trust

It was discovered that the interviewers' emphasis on the need of building trust within project teams was significant. Interviewees described that having informal meetings allows for building trust and further makes it easier to be open and ask for help and improve the internal collaboration. The majority emphasized the importance of interacting between two eyes, and that it is the most efficient way to reach out.

"You need to be open for feedback and not scared to ask for help. You don't need to know everything yourself, it is a learning process all the time." - Respondent D

Need for pull-strategy

The interviewees which elaborated and expressed most about knowledge management were also often with the perception that the capabilities and culture around knowledge sharing was great and that there is an established focus in the company.

"I try to spread the knowledge that I'm working with, so others can have it in the back of their minds. And like, oh, maybe (Respondent A) will know about this because he has done something similar." - Respondent A

But these respondents also gave a strong impression that they were the ones at the front of this area, and it could be their interest rather than something that management has pushed for them.

"So, this is the thing, I need to go after the information." - Respondent C

Not knowing what others know

The view employees had about their perception on how well they knew who possesses what expertise within the company differed. The primary factor which could be identified was project group size since knowledge rarely broke out of the project groups. This led to contrast between larger groups where knowledge about coworkers' expertise was dispersed, and smaller groups becoming isolated from inContext AB when it came down to insights about their coworkers.

"I don't know who is knowledgeable in certain areas." - Respondent C

"I know who is knowledgeable, and I feel like knowing who to ask makes my job way easier." - Respondent E

This extended to good procedures from team leaders. Like the work structure, the procedures got captivated within the group as well. Currently there are not regular movements between group members and project groups allowing for insights to spread. Worth to mention is also that some respondents expressed that they preferred being part of the same group for long periods of time.

Limitations in connection to confidentiality

The topic of confidentiality has been brought up a number of times since the start of this project. It has been mentioned in every interview, as well as unstructured conversations made while at the office. Confidentiality when working towards a company was found to present a number of problems.

"It's a problem with a company like this, we can't save the customers' stuff anyway. Every customer is unique, which makes the stuff unique. But the good thing is that we don't jump between customers anyway." - Respondent F

The clear one is that it hinders how detailed something can be explained without revealing information that is legally not allowed. But it also works as an incentive to not share information since, firstly there are clear instructions that some information cannot be shared and secondly the process of doing so becomes more complex and requires more thought. Also not wanting to be eligible for violating contracts will incentivize caution. This differs somewhat amongst the respondents, depending on their role, since one respondent did not see confidentiality as a problem.

"No, I wouldn't say. But again, like my role as a project manager, I'm following processes, not so much technical details." - Respondent H

However, inContext AB was found not to have processes of advising employees when it comes to confidentiality to help them share knowledge in a legal manner and helping them see possibilities instead of restrictions.

No incentives in connection to knowledge management

What was found from the interviews was that there are currently no incentives for knowledge-related activities such as seeking, learning, and sharing knowledge. However, there are some general incentives in place. For example, if you do good in a specific area, you will for instance be honored during the upcoming greeting.

"They can't keep giving me a bottle of champagne every year. That kind of incentive goes away because yeah, okay, I already had my bottle of champagne." - Respondent B

4.3.1 Problem description from the interviews

Knowledge is not actively shared, instead people have to seek it to find it which is distinctive for a pull system. Although there is no intentional strategy for a pull system, the system works well to an extent. Arguably, this is because they have a strong company culture, and therefore, a more social way of sharing knowledge works. However, since there is no defined system, there are components missing that could increase the pull strategies performance as well as enhance their current efforts to create a good culture. Also, with a strong focus on a social way of performing knowledge transfer, there is a neglect of maximizing the utilization of push strategies where possible. This is primarily prevalent when considering refinement and using explicit knowledge between the consultant and inContext AB, as shown in Figure 4 below.

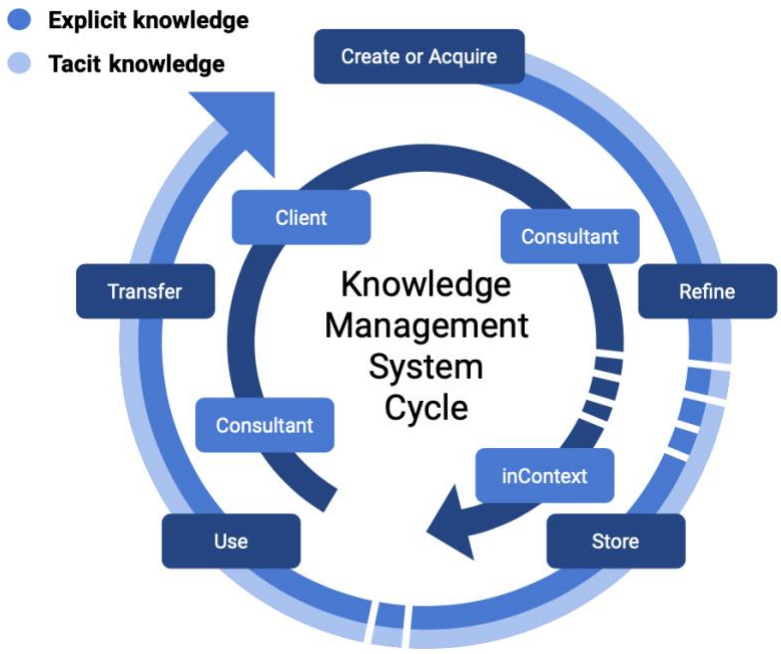


Figure 4. Visualization of the identified issue. The loops are incomplete, as seen with the disruptions, due to missing components in the cycle of KMS. This makes it difficult to share both explicit and tacit knowledge between refine, store, and use. The transfer from the consultants to inContext AB as an organization is involved in this.

4.4 Design of the KMS

In the following chapter, the results from systematic evaluation of dimensions, categories, and critical success factors are presented focusing on inContext AB internally.

Dimensions

External influence dimension was eliminated since it was deemed to have meagre impact in this case study. The organization in question is not restricted or limited by political influence as well as weather and climate, and the risk is considered too low for the topic to be interesting to research in this study.

Categories

Organizational structure was deducted by prioritization of needs. inContext AB has a clear organizational structure typical for consultancy firms in place and it is regarded as enough.

Resources were deducted because all factors included were either, currently sufficient, unclear definition or included to an extent in other factors.

The benchmarking category was combined with the performance measurement category because it was considered that the two topics were similar enough.

Reengineering was deducted due to the nature of the project being development of a new knowledge management system. This means that the project already entails reengineering to an extent depending on how radical the system change proposed will be. Since it was considered throughout the project whether the suggestions in the new system were feasible for the organizations related to the project, the decision was made to remove this category.

Top management support and leadership were deducted since management is involved in this project to a great extent and is the one interested in the result and supporting it.

Human resource management was deducted since factors included are already recognized, measured, and work is done to improve them. Not necessarily from a KM perspective, but the argument is still made to consider other categories more critical.

Knowledge creation was deducted to further increase the relevancy of the study towards consultancy firms. Since consultancy firms specialize in solving problems for different customers, knowledge creation is always something that occurs to a certain extent, hence that is rarely a significant problem.

Critical success factors deducted

Clearly articulated knowledge management strategy was deducted since well-articulated strategies, like the personal development plan, that incorporate KM goals are in place. This project report can also be argued to be considered as an extensive articulated knowledge management strategy when completed.

Communication/marketing factor was eliminated due to organizations' current good capabilities of making their employees aware of what is happening within the organization. Therefore, this factor can be considered noncritical.

Holistic approach is deducted because of time constraints for the project. A holistic approach was used in the beginning, but as the project has developed, delimitations have been necessary due to time constraints.

Since the factor KM strategy is a broad factor, it was removed. The KM strategy will be better defined as the result of the interaction of several other factors, and therefore it is selected to keep the included factors separately.

Chief knowledge officer (CKO) as a separate position is deducted due to the study involving a smaller mid-size firm. A critical trade off will therefore be to not assign an explicit separate position for the CKO role for the economic benefits, but with the trade-off of placing this responsibility on the CEO or senior managers. This is possible to do successfully, according to Tiwana (2000), and it's also stated that knowledge management should be self-destroying if it's truly successful.

For the same reason, because inContext AB is a small-midsize business, having a KM department wouldn't be financially smart and is therefore eliminated as a critical success factor in this case. This is something that could be implemented later when the company gets bigger and operates full-scale in different offices to keep control over knowledge management.

Variety of knowledge sources was deducted since inContext AB is interested in knowing how they can utilize the knowledge of their consultants more effectively and not find new ways of gaining knowledge. There is also no immediate need since obtaining knowledge to solve problems is not an issue.

Mechanism to approve activities was removed due to the nature of inContext AB as a flexible firm with a relatively flat hierarchy. Currently, there are no issues approving activities and decisions can be made by the management group without the need for long processes.

Since this is an early implementation of a KMS, the factor of precise KM processes is something that will be a critical success factor at a later stage and is therefore eliminated. The creation of a KM process is within the scope of this project, which can be further evaluated and refined to be precise.

Readiness to accept the new systems was deducted in the hope that together with management being onboard and this thorough project being executed, it would be enough to convey the new

system. Instead, this factor is considered throughout the project so that sound arguments are made as well as benefits and challenges.

To create a risk-taking climate is also deducted due for two reasons. Firstly, the general perception of risk among the employees is that major risks are rarely encountered. Secondly, the attitude towards risk-taking is that it is allowed since the risks encountered are small. Therefore, this factor falls outside the area of priorities for this project.

Managing tacit knowledge is also deducted as a factor since there was no delimitation in this project to only consider explicit knowledge. Therefore, managing tacit knowledge for different knowledge management aspects is always considered.

Common vision and goals were deducted since the review of inContext AB current capabilities revealed a well-developed personal development plan, including well-articulated goals that relate to knowledge management.

Procedural design needs to help establish a loop of knowledge-sharing was deducted as a factor since it is too broad and includes several factors itself. It can be seen as a system itself and is therefore not suitable to be included in this KMS.

4.4.1 Identified critical success factors

Identified critical success factors which are found having a significant impact on the knowledge management for inContext AB internally are the following ones:

- Alignment between KM strategy and business needs
- Collaboration
- Flexible knowledge structure
- Integration with other initiatives and work practices
- KM roles, responsibilities and leaders
- Knowledge application
- Performance measurement, KM progress tracking and measurement
- Quality information
- User-orientation

These identified critical success factors are further used to ensure that all factors are covered and considered when selecting key functions for the knowledge management system.

4.5 Validation

This chapter is presented before chapter *5. Development of the Knowledge Management System* due to wanting to present that chapter as a complete system with no new information being presented after. The open discussion about the results obtained in chapter *5. Development of the Knowledge Management System* led to a deeper understanding of each key function. The result was that no key function was invalidated, however, the prioritization and to what extent the key functions should be implemented were further understood. Formulated opinions that refined the KMS were that several clients appreciated the possibility of keeping the same consultant for longer periods of time if they perform according to or above expectations. This can be seen as an on-site consultant turnover, which has been brought up in the literature study and can therefore be supported and taken into consideration.

5 DEVELOPMENT OF THE KNOWLEDGE MANAGEMENT SYSTEM

In the following chapter the development of the framework for the knowledge management system at inContext AB will be presented.

5.1 Components of the framework

Key functions of the knowledge management system for inContext AB are shown in Table 3 below, where all included components and their category can be seen. Components already in place are only stated by name under chapters *5.5.1 - 5.5.5*, for further explanation see chapter *4.1 Infrastructural evaluation results*. The framework is inspired and structured with the SECI model in combination with the four pillars for a KMS, and SKMI. The headlines were changed to make it more general understandable. Therefore, socialization was exchanged by social sharing, externalization by enable sharing, combination by documentation, internalization by learning, and lastly SKMI was exchanged by incentives. The complete framework of the system can be found in Appendix F.

Table 3. Table of the knowledge management system key functions.

	Social sharing	Enable sharing	Documentation	Learning	Incentives
Organization	Gatherings, Mentorship				CKO, Incentives
Process	Quality assurance	Consultant profiles	Knowledge mapping		
Strategy	Technology brokering	Knowledge leaders		Try new roles, Training programs	Measurement & benchmarking, Personal development plan
System	Workplace	Metadata management	Google Drive	Learning management system	

When developing a KMS for a consultancy firm, the primary critical trade-off compared to other industries is applying a client-centric approach both in terms of capturing, storing, and sharing knowledge and in terms of security and confidentiality. inContext AB also invests a great deal of resources into the company culture and therefore a reasonable additional trade-off in this case would be to further emphasize this by designing the overall system with a pull-strategy focus.

5.1.1 Social sharing

The following subchapter presents and describes key functions for supporting tacit to tacit knowledge in a social setting.

Key functions that are already in place for supporting social sharing at inContext are:

- Gatherings
- Mentorship
- Workplace

Recommended key functions to implement are described below.

Quality assurance

Quality assurance is a current capability described previously in chapter *4.1 Infrastructural evaluation results*. It is important to capitalize on good current practices and spread them across the organization. Performing quality assurance is also an exemplary situation to identify and collect lessons learned that are valuable information of high quality for inContext AB for further knowledge application, which are two critical success factors. It's important to provide tools for easy documentation and sharing of lessons learned.

Technology brokering

Technology is a synonym to cross pollination which is a term used by IDEO. IDEO is one of the most successful design consultancy firms in the world and according to Hargadon and Sutton, (1997) acting as technology brokers has contributed to that success. The strategy IDEO uses amplifies the number of new problems designers face by firstly not specializing designers to industries and instead moving them around after completion of a single project. Designers also transfer off and on long term projects to prevent "burn out" and a designer is never on the same project for more than a year. Together with this, teams are formed around individual projects and extra designers are allocated for brainstorming sessions or short bursts of effort. This exposes designers to new knowledge as well as understanding the knowledge of their co-workers. If a problem occurs, the topic is shared openly within the company at a joint Monday meeting, allowing any designer who feels they possess knowledge to get involved. It is also expressed that this strategy works to a large extent because of the norms and attributes that IDEO considered important. A designer should not be too arrogant or insecure to ask for help and wise enough to understand that they know things others don't, therefore constantly offering their help and advice. To support this further, their reward structure is also largely based on informal and formal metrics in the form of informal reputation and formal peer reviews. In other words, it's also described as based on the respect you have among your peers, and respect is obtained from individual efforts that help produce good designs as well as helping others. Tailoring this approach to inContext AB the recommendation would be strategizing movement between industries to allow for knowledge about expertise to disperse, but adopting a longer timeframe to fulfil the wants of their employees as well as their clients. Secondly, to incorporate more formal peer reviews to further bring clarity to what management values.

5.1.2 Enable sharing

The following subchapter presents and describes key functions for supporting tacit to explicit knowledge.

Knowledge leaders

Having assigned KM roles, responsibilities, and leaders within inContext AB will mentor and develop the skills of the junior consultants. These leaders will efficiently provide valuable guidance and directions based on their experiences. By actively promoting the KM projects, its adoption, and use, will increase the awareness of the value of it. This, not only on an organizational level, but also to show what's in it for the consultants themselves. It will be easier for the

management to introduce a more pull-strategy focus, which was found needed from the interviews, by being supported by assigned KM roles, responsibilities, and leaders to push these questions to the whole organization. Further, these leaders will also make the mapping of the knowledge possessed at inContext AB easier, as well as linking it. With this, knowledge communities will occur, which further supports a culture of continuous learning and development, making the organization more competitive with an even deeper understanding of the functional area.

Consultant profiles

Consultant profiles are a way to display qualifications, certifications, and professional accomplishments. This could be combined with an internal searchable system to allow consultants to find information about their colleges expertise. The barrier to entry for this kind of system would most likely require a larger financial investment from the organization and more time and commitment from all consultants. Consultant profiles also provide benefits like improved brand ability for clients, increased business opportunities, building credibility, and networking opportunities for the consultants themselves. These are all important business needs, and an important factor for successful KM is alignment between KM strategy and business needs.

Metadata management

To ensure data loss prevention and help the organization retrieve information when needed, the implementation of metadata management is important to consider in future investments. Metadata management features allow users to classify knowledge artifacts according to relevant tags or labels, which can help organize and retrieve knowledge more effectively. Taxonomies can also provide a common language and conceptual framework for knowledge sharing across different departments or domains. For inContext AB it would be recommended to build on their current systems Google Drive and implement Drive labels, which is designed to classify content, apply policy items, and find files faster. These features make it possible to sort documents by more than names, for example date, persons, field, draft, awaiting signature, etc. Requirements for these features are Google Workplace business standard.

5.1.3 Documentation

The following subchapter presents and describes key functions for supporting explicit to explicit knowledge.

Key functions that are already in place for supporting documentation at inContext are:

- Google Drive

Recommended key functions to implement are described below.

Knowledge map

Identified from the interviews was that it is a lack of knowledge about what other consultants are working on and possess knowledge of. This is especially helpful for the consultants who have not been within the organization for a longer time or have a manager position. Therefore, a knowledge map needs to be introduced. This will visually represent the employee's knowledge and expertise, which will allow for an easier and faster connection with the right person when help or advice is needed. Having this within the KMS will also facilitate collaboration between project teams and, with that, enable knowledge sharing across the organization. Further, it will help inContext AB optimize its resources, which improves efficiency and effectiveness.

5.1.4 Learning

The following subchapter presents and describes key functions for supporting explicit to tacit knowledge.

Key functions that are already in place for supporting learning at inContext are:

- Training programs
- Try new roles

Recommended key functions to implement are described below.

Learning management system

A learning management system (LMS) is a platform designed for an organization to be able to share learning material in an interactive way. A challenge identified at inContext AB was that learning material has been forgotten or left unused even though efforts have been made to construct it. According to Rosenstiel (2011) humans interact based on four principles: situational enabling, empowerment and obligation, and individual desire. The current way of sharing learning material could be improved in situational enabling and empowerment by making the content more interactive with the help of an LMS. There, all learning material could be collected and organized as courses with progression and completion. A study by Friedrich et al. (2019) showed that gamification elements can be incorporated to further motivate people, and this could be combined with an LMS. With the creation of profiles, contributions, ranking, personal progress, levels can be tracked and achievements, badges, rewards, feedback can be given.

5.1.5 Incentives

The following subchapter presents and describes incentives to support knowledge management.

Key functions that are already in place to support knowledge management at inContext are:

- Personal development plan

Recommended key functions to implement are described below.

Incentives

To create and uphold a KMS, participation and commitment play an important role. Incentives need to be carefully thought out to effectively help solve the problem that exists. The qualitative study conducted in this project shows that there is a lack of helpful explicit knowledge documents relating to customer organizations, standardized ways of working, good practices, etc. To promote a push strategy in these areas, an incentive can be incorporated to promote the sharing of knowledge. A study by Govind and Suryanarayanan (2013), shows that ambiguity tolerant individuals, which is arguably more common in consultants, considered “an incentive mechanism that rewards the contributor for shared knowledge used by a knowledge user, and the knowledge user for the act of reuse, is more effective than a simpler incentive scheme that merely rewards knowledge sharing when usefulness level is low”. Therefore, rewarding how much an individual shares should be combined with rewarding its usefulness. It was also shown that the usefulness is as important as the incentive. Due to that, usefulness and incentives need to be increased in a concentrated manner to ensure quality information.

Measurements and benchmarking

To ensure the maturing of the KMS, it is essential to measure the system's impact and outcome. There are different methodologies for measuring knowledge management. Worth mentioning are Wibowo and Grandhi (2016), who present a fuzzy multicriteria group decision approach for small to mid-size firms. The approach can be regarded as complex and requires someone to be educated in it. It also requires involvement from the entire management group, but the method handles the complexity of evaluating KM as it involves several decision-makers, multiple evaluation criteria, and the presence of subjective and imprecise assessments in the group decision-making process. Another method would be the user satisfaction-based measurement method presented by Lo and Chin (2006). The trade-off would be that the method is less refined and more up for interpretation as the specific assessment questions are not defined. The benefit would be not neglecting inContext AB current capabilities of measuring user satisfaction in other areas. Therefore, arguably this methodology is easier to incorporate.

Chief knowledge officer

The responsibility of the CKO would in this case fall on senior management. The focus would be according to Tiwana, (2000) on "correcting the knowledge flow and eliminating related deficiencies and inefficiencies that exist within the company (chpt.13)". In the same chapter, the job description of a CKO is stated, which has been rephrased and tailored to inContext AB and is stated below. The job description of the manager with assigned CKO responsibilities would be:

1. Optimizing process design for knowledge management: Improving processes for creating new knowledge, distributing existing knowledge, and applying or reusing what is already known by implementing suggested processes. Also, improving current capabilities and processes.
2. Implementing new channels: Implementing suggested channels for leveraging untapped knowledge and competencies within the firm, such as knowledge mapping and consultant profiles.
3. Integrating knowledge management: Embody knowledge management in the routine tasks and activities of the firm's employees. Distribute the knowledge of this project to team leaders and emphasize its value.
4. Breaking barriers and eliminating impediments: Break down technical barriers in communication and knowledge exchange processes with a socializing approach to sharing knowledge in mind.
5. Watching the learning loop: Ensure that the firm is learning from its past mistakes and failures. This also applies to the KMS.

6 CONCLUSIONS AND DISCUSSION

The following chapter presents the conclusions of the project and a discussion together with recommended future work within this field are presented.

6.1 Conclusions

The study highlights the importance of understanding which critical factors prevent and encourage knowledge sharing. The answer to the first research question of this study, as presented in the results section, is that the following nine factors are the most critical ones:

- Alignment between KM strategy and business needs
- Collaboration
- Flexible knowledge structure
    - Which is being allowed and having the possibility to retrieve knowledge from other contexts and apply it to new contexts.
- Integration with other initiatives and work practices
- KM roles, responsibilities and leaders
- Knowledge application
    - It is defined as use and application of available knowledge to make decisions and perform tasks through direction and routines.
- Performance measurement, KM progress tracking and measurement
- Quality information
    - Is ensuring that information provided follows a high standard.

When having these critical success factors identified, further investigations in which key functions needs to be included in the knowledge management system could be done.

The findings for the second research question, key functions in a knowledge management system, are the following ones:

- For tacit to tacit knowledge sharing:
    - Gatherings
    - Mentorship
    - Quality assurance
    - Technology brokering
    - Workplace
- For tacit to explicit knowledge sharing:
    - Consultant profiles
    - Knowledge leaders
    - Metadata management
- For explicit to explicit knowledge sharing:
    - Google Drive
    - Knowledge mapping
- For explicit to tacit knowledge sharing:
    - Learning management system
    - Training programs
    - Try new roles
- Incentives for knowledge sharing:
    - Chief knowledge officer
    - Incentives
    - Measurement & benchmarking
    - Personal development plan

The nine critical success factors are all covered and considered by the suggested key functions included in the developed KMS framework, and the framework can therefore be seen as a relevant recommendation for a consultancy company like inContext AB.

Furthermore, critical trade-offs that must be considered when developing such a system were found to be budget, effort, and implementation time. Depending on the size of the company, there will, for natural reasons, be differences in how much money can be invested in the KMS. A large company can implement a KMS without getting financially affected, whether it works or not, whereas a smaller company runs a large risk if substantial investment has been put into the system without receiving anything out of it. The same goes for the amount of effort that can be placed into the KMS. Implementation time is also a trade-off that needs to be considered when developing the KMS. If the KMS is focused on prioritizing short-term effects, it may be found to be more responsive and successful for natural reasons since the results will be seen faster. On the other hand, such a prioritization may overlook long-term strategic objectives. What can be seen is that a KMS that focuses on long-term strategic goals may be more sustainable and powerful but less responsive to urgent demands. The risk with a long-term focus is that it gets deprioritized as a result of being interpreted as taking too much effort.

When presenting the developed knowledge management system, it was apparent, as predicted, that all key functions could not be implemented simultaneously. Therefore, there is a need to prioritize which key function to start with. Based on the dialogue that took place in the open discussion about the system as well as logical reasoning, a sound way to implement the system would be to start by standardizing already existing key functions such as quality assurance. Thereafter, implement key functions that build and further develop current systems, such as metadata management, formal peer reviews, and incentives. The last step would be to implement new key functions, and we would argue that consulting profiles or a LMS would be the most beneficial investment in regard to the current challenges that inContext AB faces.

6.2 Discussion

One identified challenge to consider is client confidentiality. Clients in these industries that inContext AB works in typically use non-disclosure agreements with their employers. These agreements place strict restrictions on who can access and share confidential information, as well as how it can be used and under what circumstances knowledge can be shared. However, the good thing is that these businesses tend to have a limited number of clients, and by that a significant part of the organization is involved in projects for the same client. This means that those working on the same client project can benefit from a strong knowledge-sharing culture. This arrangement, however, can have an impact on the consultancy firm because it creates gaps within the organization and hinders critical success factors and key functions of the KMS by limiting knowledge sharing throughout the entire organization. By carefully managing the tension between client confidentiality and internal knowledge sharing, smaller companies can however maximize the benefits of an implemented KMS, enabling efficient collaboration between different clients and leveraging collective expertise while remaining loyal to the obligations imposed by non-disclosure agreements.

Another challenge with the KMS that a smaller company has a great possibility of handling is related to the consultants soft skills. What has been noticed is that soft skills play an important role in the consultancy domain, particularly when it comes to executing customer-based projects. In this realm, effective collaboration and interaction with project partners, managers, and customers emerge as critical requirements. According to Castrillon (2022) soft skills are personal attributes, attitudes, and behaviors that are more difficult to learn in a traditional way. Castrillon (2022) further describes their enduring nature, which, on the other hand, is noteworthy, as once acquired, these skills remain with individuals throughout their careers. Soft skill development eliminates the

need for employer-provided training, increasing an individual's desirability in professional settings. The benefit for a smaller company is that they can more easily employ consultants who have the right soft skills since they do not have to hire a large number of consultants, which allows them to be selective when choosing.

The SECI model was found to be appropriate as a template for the framework's development because it visualizes the continuous loop of knowledge-sharing that is desired in a KMS, as well as the various stages of transfer between the two different types of knowledge. However, there are some limitations with the SECI model that must be considered. It can be perceived as oversimplifying the complexity of knowledge creation and sharing. This is because it assumes that knowledge is easily shared and that everyone involved has the same access to it. According to Lin (2007), the model fails to take into account power dynamics, organizational culture, and individual motivation factors for establishing knowledge sharing behaviors. However, in this project, those have been considered critical factors. What was found from the study is that inContext AB has the opportunity to tackle the skepticism about those factors. Furthermore, the article by Lin (2007) stresses that the SECI model provides no clear guidelines for how to implement the practices in the organization. Regarding the argument, the model has been used in this project as a component when building the system and integrating the four dimensions to provide the desired loop of knowledge from tacit to explicit and vice versa. Therefore, the lack of guidelines for implementation does not make the usage of the model vague in this project.

Interviews were chosen since the company is relatively small, and therefore the insights from the consultants were efficiently obtained through interviews and observations. If surveys had been used, the study might have included more consultants. Arguably, the data that was useful for our study was qualitative rather than quantitative, and a deeper understanding is desired. Surveys were not used because the data would not have been as detailed and well-supported otherwise.

Due to the project being carried out over a 20-week period, the inevitable constraint was not being able to validate if the proposed system was beneficial to knowledge management. A KMS is, as mentioned earlier, interactive to a large extent and needs participation and commitment. Measuring the performance of a KMS is also a complex task to carry out. Therefore, evaluation and validation of a KMS would take in the vicinity of a year to perform. Nevertheless, the entire project is based on strong academic literature studies, qualitative research, and scientific methodologies, which give the proposed system a strong chance of success.

The main limitation of this thesis was that the case study data was collected from one consultancy firm. This poses a variety of limitations, such as its applicability to other consultancy firms. The key functions proposed can be beneficial to any consultancy firm, but each firm would need to perform their own infrastructural evaluation and evaluate their own capabilities in terms of need, time, and budget before choosing key functions. It is also recognized that the size and scattering of a firm can influence the infrastructure evaluation and, therefore, the necessary key functions. Therefore, while developing the system, the expansion and growth of the firm were considered, even though they may not have been phrased specifically.

6.2.1 Recommended future work

Three recommendations for future work are provided below.

Measure the effects of the key functions

Measuring the effects of each key function within the KMS would be useful for future work. This requires a systematic way to measure their effects, which might be accomplished by establishing key performance indicators (KPI) that fit with the objectives of each important function within the KMS. These KPI should provide measurable criteria for evaluating the effectiveness of each key

function. For example, in the key functions of knowledge mapping and knowledge leaders, where the goal is to spread awareness of the consultant's knowledge and experience, relevant KPIs could include the number of successful searches answered or the time required to locate relevant knowledge owners. Additional adjustments and iterations based on the insights gathered from these KPIs can be used to improve the KMS.

Cost estimate

Implementation of the suggested KMS entails both the required initial investment and ongoing system costs. As a result, a comprehensive cost estimation would be useful and therefore a suggestion for future work. Identifying initial costs would be brought up in such an estimation. This would determine the cost of KMS software licenses, implementation services, and costs in connection with infrastructure changes, among others. In addition, ongoing costs would have to be considered, which include costs for the personnel and resources needed to manage and maintain the KMS. Another cost is the one associated with integrating the KMS with existing systems and potential future upgrades, which should also be considered. These ongoing expenses are examples of what contributes to the total cost of operating the KMS over its lifetime. In addition, calculating potential cost savings and determining the Return on Investment (ROI) can provide valuable insights for the management team. This is to evaluate the financial benefits of implementing the KMS by comparing the identified costs to the estimated cost savings. Future work and analysis like this would serve as a baseline for evaluating how well the suggested system performs in terms of improved business as a result of improved access to critical knowledge.

Assessment of developed framework

This master's thesis presents a new way of ensuring that a developed KMS is complete by evaluating it against a new framework that combines the SECI with the four pillars of knowledge management as well as considering SKMIs. The recommended future work could therefore be to validate if this method is good and suitable to apply to all organizations by conducting comprehensive case studies across different industries and organizational settings to provide valuable insights into the feasibility and adaptability of the proposed framework. This is to analyze and identify any potential weaknesses and further improvements.

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Interview guide

Introduction questions:

1. Kan du beskriva din arbetsroll?
   Can you describe your role at inContext?

2. Hur länge har du varit på inContext?
   a. Hur många projekt?
   How long have you been with inContext?
   a. *How many projects?*

3. Kan du berätta om dina erfarenheter inom konsultbranschen?
   a. Hur lång tid har du jobbat inom branschen?
   b. Olika positioner inom branschen?
   Can you tell us about your experience in the consulting industry?
   a. *How long have you worked in the industry?*
   b. *What different positions have you held within the industry?*

4. Vilket projekt jobbar du med och hur ser projektgruppen ut?
   a. Hur många andra projektgrupper ingår i det projektet?
   What project are you currently working on, and what does the project team look like?
   a. *How many other project teams are involved in that project?*

5. Tycker du att KM är viktigt i dagens affärsklimat?
   a. I så fall, varför?
   Do you believe that knowledge management is important in today's business climate?
   a. *If so, why?*

Focus questions (project):

6. Är kunskap från tidigare projekt en viktig del för att kunna hantera nya projekt på inContext?
   Is knowledge from previous projects or other colleagues an important part of being able to handle new projects at inContext?

7. Vilka kunskapshantering åtgärder sätter du/ni upp vid start av ett projekt?
   What KM actions do you set up at the start of a new project?

8. Vilka kunskapsdelning åtgärder utför du/ni under projektets gång?
   What knowledge-sharing actions do you carry out during the project?

9. Vilka kunskapsdelning åtgärder utför du/ni efter ett projekt?
   What knowledge-sharing actions do you carry out after a project?

Focus questions (individual level):

10. Hur arbetar du idag för att fånga in kunskap och lära dig nya saker på inContext?
    How do you currently work to capture knowledge and learn new things at inContext?

11. Hur arbetar du idag för att dela med dig av din kunskap till andra på inContext?
    a. Vilka delar du din kunskap till?

    How do you currently work to share your knowledge with others at inContext?
    a. *Who do you share your knowledge with?*

12. Hur arbetar du idag för att lagra din kunskap till framtida projekt eller framtida medarbetare?
    How do you currently work to store your knowledge for future projects or future colleagues?

13. Utifrån din erfarenhet, i vilket sammanhang och hur sker den bästa kunskapsöverföringen?
    Based on your experience, in what context and how does the best knowledge transfer occur?

14. Ser du någon nackdel med att dela med dig av den kunskap och expertis du besitter?
    Do you see any disadvantages to sharing the knowledge and expertise you possess?

Focus questions (structure):

15. Känner du att du har bra koll på vilka som sitter på kunskap inom ett visst område?
    If you need some input, do you know who is knowledgeable in a certain area?

16. Anser du att du har tydliga expertisområden som du ansvarar för?
    Do you consider that you have areas of expertise that you are responsible for?

17. Anser du att du är väl bekant med inContext personliga utvecklingsplan?
    a. Om ja, till vilken grad har du den i åtanke mellan de årliga utvecklingssamtalen?
    b. Om ja, känner du att du blir tillräckligt utmärkt för hur du presterar enligt de uppsatta målen?
    c. Om inte, varför inte?

    Do you feel that you are familiar with inContext's personal development plan?
    a. *If so, to what extent do you keep it in mind during annual performance reviews?*
    b. *If so, do you feel that you are adequately recognized for how you perform against the set goals?*
    c. *If not, why not?*

18. Har du ett sekretessavtal på det projekt du jobbar med just nu?
    Do you have a confidentiality agreement on the project you are currently working on?

19. Anser du att du har bra koll på vilken information du får dela enligt ditt sekretessavtal?
    Do you feel that you have a good understanding of what information you can share according to your confidentiality agreement?

20. Till vilken grad anser du att sekretessavtalet hämmar din vilja att dela information till inContext?
    To what extent do you feel that the confidentiality agreement hinders your willingness to share information with inContext?

Focus questions (culture):

21. Hur anser du att inContext ser på risktagande och lärande av att misslyckas?
    How do you feel that inContext views risk-taking and "learning by failing"?

22. Utförs några teambuilding aktiviteter i början av projekt för att bygga till exempel, förtroende, öppenhet, samarbetsvilja?
    Are any team-building activities carried out at the beginning of the project? To build for example trust, openness, cooperation, etc.

23. Vilka personliga förmågor tror du är viktiga för en lyckad kunskapshantering?
    a. I så fall, har du utvecklat dessa förmågor under din karriär?
    What personal skills do you believe are important for successful knowledge management?
    a. *If so, have you developed these skills during your time at inContext?*

24. Hur bidrar du till en kultur som uppmuntrar samarbete och delning av kunskap inom konsultbranschen?
    How do you contribute to a culture that encourages collaboration and knowledge-sharing in the consulting industry?

Focus questions (strategy):

25. Har du identifierat kritiska kunskapsluckor inom organisationen?
    a. I så fall, hur skulle du hantera dem?
    Have you identified critical knowledge gaps within the organization?
    a. *If so, how would you address them?*

26. Anser du att du blir utvärderad och belönad efter kunskapsrelaterade områden, som sökandet, lärandet och delandet av kunskap?
    a. Hur gynnas du av att vara bra i dessa områden?
    Do you feel that you are evaluated based on knowledge-related areas, such as seeking, learning, and sharing knowledge?
    a. How do you benefit from being good in these areas?

APPENDIX B:4: CSF - ORGANIZATION DIMENSION

Table 4. Critical Success Factors of KM: Organization Dimension. (Onofre & Teixeira, 2022b)

Category	Critical Success Factors
KM Strategy	**1.** Alignment between KM strategy and business needs; **2.** Clearly articulated KM Strategy; **3.** Communication and Marketing; **4.** Holistic approach; **5.** KM Strategy; **6.** KM Value Proposition; **7.** Integration with other initiatives and work practices; **8.** Strategic Planning; **9.** Pilot; **10.** User Orientation; **11.** Enterprise-wide and business unit specific needs
Organizational Structure	**12.** Organizational structure; **13.** Flat structure; **14.** Flexible structure; **15.** Formalization; **16.** Centralization; **17.** Decentralization
Performance Measurement	**18.** Performance Measurement; **19.** Financial performance; **20.** Non-financial performance; **21.** KM progress tracking and measurement; **22.** Business performance
Resources	**23.** Availability of resources; **24.** Financial resources; **25.** Free time and space; **26.** Use of consultants; **27.** Customer and knowledge supplier relationships
KM Organization	**28.** Chief Knowledge Officer; **29.** KM Champions and Leaders; **30.** KM Department; **31.** Specialized KM team; **32.** KM roles and responsibilities; **33.** Clearly defined knowledge ownership; **34.** Knowledge communities; **35.** Network of experts
Benchmarking	**36.** Benchmarking
Reengineering	**37.** Reengineering
Organizational Culture	**38.** Organizational Culture; **39.** Collaboration; **40.** Trust; **41.** Common vision and goals; **42.** Knowledge creating and sharing culture; **43.** Knowledge-centered culture; **44.** Learning culture; **45.** Learning from failure; **46.** Teamwork; **47.** Transparency; **48.** Openness; **49.** Readiness to accept the new system; **50.** Risk-taking climate
Top Management Support and Leadership	**51.** Top management support; **52.** Leadership
Training	**53.** Training
Human Resource Management	**54.** Human Resource Management; **55.** Human Resources; **56.** Employee commitment; **57.** Employee empowerment; **58.** Employee involvement; **59.** Employee motivation; **60.** Employee retention; **61.** Human capital; **62.** Job security
Rewards	**63.** Incentives and rewards

APPENDIX C:5: CSF - TECHNOLOGY DIMENSION

Table 5. Critical Success Factors of KM: Technology Dimension. (Onofre & Teixeira, 2022b)

Category	Critical Success Factors
IT Application	**64.** IT Application; **65.** Balance between people and IT; **66.** Technology; **67.** Learner-focused technology; **68.** Alignment between business and technology; **69.** Friendly and easy to use KM system; **70.** KM system design
IT Infrastructure	**71.** Effective IT infrastructure; **72;** Access to network infrastructure and hardware; **73.** Security
Technology tools	**74.** Collaborative tools; **75.** Effective KM tools; **76.** Functions of KMS; **77.** Knowledge repository

APPENDIX D:6: CSF - KNOWLEDGE AND KM CAPABILITY DIMENSION

Table 6. Critical Success Factors of KM: Knowledge and KM Capability Dimension. (Onofre & Teixeira, 2022b)

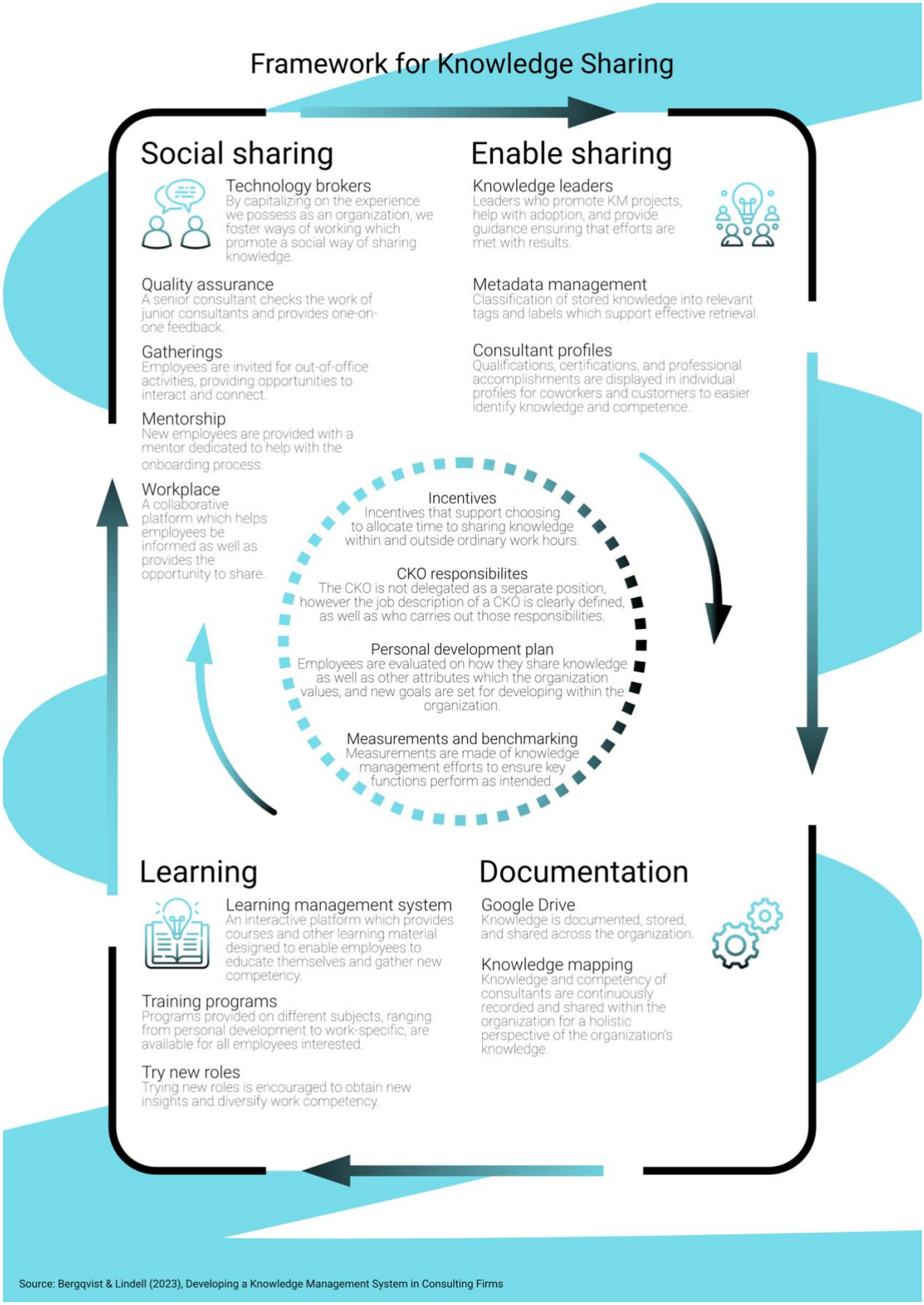
Category	Critical Success Factors
Knowledge Structure and Quality	**78.** Flexible knowledge structure; **79.** Knowledge structure and map; **80.** Knowledge architecture; **81.** Nature of knowledge; **82.** Quality of information; **83.** Variety of knowledge sources
Knowledge Process Management	**84.** KM processes and procedures; **85.** KM processes easy to follow; **86.** Knowledge process management; **87.**Managing explicit and tacit knowledge; **88.** Managing knowledge throughout its lifecycle; **89.** Mechanism to approve activities; **90.** Precise KM processes
Knowledge Creation	**91.** Knowledge Creation; **92.** Innovation
Knowledge Capture and Storage	**93.** Knowledge capture; **94.** Knowledge storage; **95.** Knowledge identification
Knowledge Sharing	**96.** Knowledge Sharing; **97.** Multiple channels for knowledge sharing; **98.** Specialized meetings, conferences and seminars; **99.** Procedural design needs to help to establish a loop of knowledge-sharing
Knowledge Application	**100.** Knowledge Application

APPENDIX E:7: CSF - EXTERNAL INFLUENCE DIMENSION

Table 7. Critical Success Factors of KM: External Influence Dimension. (Onofre & Teixeira, 2022b)

Category	Critical Success Factors
Socio-economic Environment	**101.** Socio-economic environment; **102.** Economic climates; **103.** Economic stability; **104.** People affected because of the project activities; **105.** People benefiting from the project; **106.** Sound economic policy; **107.** Surrounding neighbours affected because of the project activities
Political Influence	**108.** Sources of finance; **109.** Confidence of politicians; **110.** Regulations; **111.** Adaptability to amendment of project plans
Industry Influence	**112.** Availability of external resources; **113.** Subcontractors and suppliers; **114.** Market prices of materials and labour
Environmental Influence	**115.** Ground conditions of projects; **116.** Weather conditions

APPENDIX F:8: FRAMEWORK FOR KNOWLEDGE SHARING



Source: Bergqvist & Lindell (2023), Developing a Knowledge Management System in Consulting Firms

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